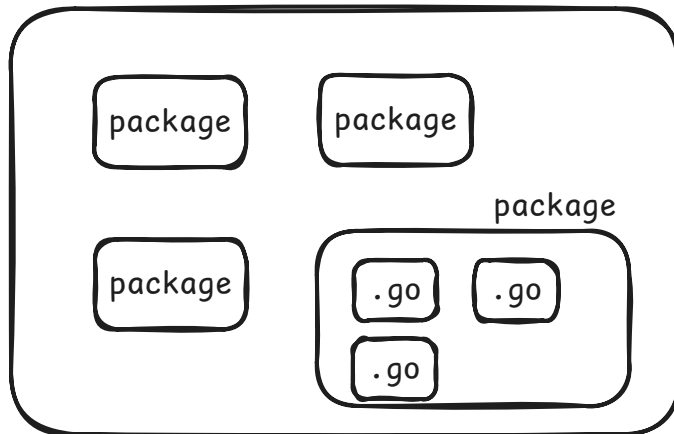


Lecture 01 - Go Foundations

```
go mod init <name>
```

```
go mod init github.com/salikhP/todo-app
```

Module/Repository



go.mod

go.sum

```
go run .      go run main.go
```

```
go build .
```

```
go fmt .
```

```
go test .
```

```
go mod tidy
```

```
name := Aibek
age := 25
height := 180.0
```

```
var (
    name string
    age int
    height float64
)
```

```
var age int = 25
var age = 25
age := 25
```

int64 int32

<type>: int
<name>: age

25

Don't try to make the Go code as short as possible.
Try to make it obvious.

Zero values

```
var name string  
fmt.Println(name)
```



" "

None, null, nil

```
type User struct {  
    IsAdmin bool  
}
```

Constants

represents a value that can't be reassigned

```
const Pi = 3.14.....  
const limits = 500...
```

```
const x = 10
```

```
var a int = x  
var b float64 = x  
var c int32 = x
```

Type conversion

```
var x int = 10  
var y int32 = 20
```

```
x = y
```

```
x = int(y)
```

<type>: int
<name>: age

10

<type>: int32
<name>: age

20

Integers

```
a := 5
```

```
b := 2
```

```
fmt.Println(a / b) // 2
```

```
2.5
```

```
fmt.Println(float64(a) / float64(b))
```

Garbage Collector

```
func hello() string {  
    name := "Dulat"  
    return name  
}
```

```
func main() {  
    fmt.Println("Hi!")  
}
```

```
func hello(name string) string {}
```

```
func hello(name string) string, int {}
```

```
func hello(name string) {}
```

